

Aidan Truel

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Education

MSc in Neural Systems and Computation — UZH and ETH Zürich	GPA 5.6/6.0	09/2024 — 10/2026
B.S. in Mathematics - Computer Science — UC San Diego	GPA 3.85/4.00	09/2020 — 03/2024
B.S. in Cognitive Science - Machine Learning — UC San Diego	GPA 3.85/4.00	09/2020 — 03/2024

Experience

[Graduate Research Assistant](#) — Institute of Neuroinformatics September 2024 — Present

- Master's thesis with Professor Timothée Proix applying methods from control theory and nonlinear dynamics to understand the neural basis of speech perception
- Research project with Professor Timothée Proix to investigate phoneme representations during natural speech perception by decoding phonetic features from intracranial EEG recordings
- Research project with Professor Richard Hahnloser examining functional connectivity between Broca's Area and Premotor Cortex during speech production

[Undergraduate Research Assistant](#) — de Sa Natural Computation Lab October 2022 — May 2024

- Developed novel architectures to improve the performance and usability of brain-computer interfaces, improving clinical viability
- Collected EEG data and set up real-time processing pipelines for brain-computer interface experiments

[Instructional Assistant](#) — UCSD Cognitive Science Department September 2022 — March 2024

- Taught COGS 11: Minds and Brains with Professor Mary Boyle
- Taught COGS 118C: Neural Signal Processing with Professor Eran Mukamel
- Led weekly discussion sections with 50+ students
- Created and graded weekly quizzes and assignments

[Project Lead](#) — Triton Neurotech at UCSD September 2022 — June 2023

- Led a team of ten to use convolutional neural networks and generative AI (diffusion models) to reconstruct images from fMRI data during visual perception
- Led a team to classify hand gestures from a custom-made EMG-wristband for real time computer control, which won second place at the international NeurotechX Student Clubs Competition in June 2023
- Taught workshops on neural data analysis to club members

[Software Engineering Intern](#) — Ready.net June 2022 — September 2022

- Generated documentation and tests for the UI, Apollo Graph QL resolvers, and all backend utilities
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Projects

Phoneme Representations in Natural Speech

- Investigated the spatial and temporal distribution of phonetic information across cortex during naturalistic speech perception, using single channel SEEG decoding
- Accepted as a poster presentation at the 2026 Society for the Neurobiology of Language Conference

Neural Basis of Speech Production

- Examined functional connectivity between Broca's Area and Premotor Cortex during speech production with multivariate Granger causality analysis
- Discovered feedforward information flow from dorsal BA44 to BA6v and feedback from ventral BA6v to BA44, and a striking dissociation between word and sentence production

Zinc Dry Electrodes for Wearable Electrophysiology

- Characterized novel concentric-ring electrodes with electrochemical impedance spectroscopy and on-body EMG testing, showing the suitability of zinc as a material for electrophysiology
- Presented at the IEEE Engineering in Medicine and Biology Conference in June, 2024

EEG Handwriting BCI

- Collected EEG data and pen-tip trajectories while a healthy subject wrote individual characters on a tablet
- Leveraged EEGNet to decode pen velocity from neural activity, leading to character classification with 61% accuracy

Aeromus

- Created an EMG-enabled computer mouse to navigate a laptop in real-time with hand gestures and air scrolling
- Awarded second place overall at the international NeurotechX Student Clubs Competition and won the Neuroethics competition

Pretraining Handwriting BCI

- Pre-trained an RNN on large handwriting datasets to decode text from the neural representation of attempted handwriting in a locked-in patient, leading to a 1.5% reduction in character error rate
- Developed a multi-subject pretraining procedure that overweights training data from subjects whose handwriting style most closely matches the test subject, improving cross-subject generalization

Spelling BCI

- Developed a hybrid motor-imagery and 10-class SSVEP brain-computer interface to play the New York Times Spelling Bee game in real time using OpenBCI hardware
 - Achieved 70–95% accuracy across subjects
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Skills

Programming — Python, MATLAB, R, SQL, Bash, git, Linux, SLURM

Libraries — MNE-Python, scikit-learn, PyTorch, PsychoPy, PyLSL, EEGLab, ERPLab, NLTK, spaCy, Hugging Face Transformers

Neural — Electrophysiology (EEG, sEEG, ECOG), Preprocessing (rereferencing, artifact rejection, filtering, epoching), ERP Analysis, Neural Encoding and Decoding Models, Functional Connectivity, Source Localization

Modeling — Time Series (spectral analysis, time-frequency decomposition, phase-amplitude coupling), Dimensionality Reduction (PCA, ICA, CSP), Statistical Inference (cluster-based permutation testing, Bayesian methods), Deep Learning (EEGNet, CNNs, RNNs, transformers)

Languages — English (native), Spanish (B2), German (B1)

Relevant Coursework

Applied Math — Differential Equations, Probability Theory, Stochastic Processes, Statistics, Recursive Estimation, Convex Optimization, Signal Processing, Network Science

Computer Science — Advanced Data Structures, Analysis of Algorithms, Software Tools and Techniques

Machine Learning — Machine Learning for EEG/BCIs, Probabilistic Reasoning and Learning, Learning in Deep Artificial and Biological Neural Networks, Computer Vision

Neuroscience — Cognitive Neuroscience, Neuroinformatics, Translational Neuromodeling, Neural Signal Processing, Neural Control of Movement, Computational Vision

Language — Deep Learning for Natural Language Understanding, Neural Foundation of Speech, Evolutionary Language Science

Awards & Achievements

NSF Research Experience in Interdisciplinary Artificial Intelligence 2023-24

Received a prestigious scholarship to support a year-long research project and training in artificial intelligence.

HDSI Undergraduate Scholarship Recipient 2023-24

Awarded a scholarship offering funds and mentorship for student-led data science projects.

NSF REU Summer 2023

Received funds to work full-time with Professor Virginia de Sa on testing novel concentric ring electrodes for use in motor imagery EEG studies and BCI applications.

Second Place at NeuroTechX Student Clubs Competition 2023

International competition among collegiate neurotechnology clubs. Won second place and the neurotech ethics prize for the [Aeromus](#) project.

Sixth College Provost Honors 2020-2024

USA Triathlon Collegiate National Competitor

Eagle Scout Award

Presentations

Society for the Neurobiology of Language Conference 2026

IEEE Engineering in Medicine and Biology Conference 2024

California Neurotechnology Symposium 2024

California Neurotechnology Symposium 2023

References

Prof. Timothée Proix

Assistant Professor

Institute of Neuroinformatics

UZH and ETH Zürich

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Master's thesis supervisor

Prof. Virginia de Sa

Professor of Cognitive Science

Department of Cognitive Science

UC San Diego

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Undergraduate research supervisor